

Lesson Plan

Name of the Programme:	B.Tech(H)	
Title of the Course:	Ethical Hacking Essentials	
Course code	CS3432	
Semester:	VI	
Names of the Course Instructor(s)	Dr Basavaraj Patil; Mr Rohith Raju	
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
2:0:2	30+0+30	60

Session	Module No.	Topic	Pedagogy / Activities	Pre-reading / Reference
1	M1	Introduction to Cyber Security & Cyber Crimes	Lecture, real-world breach discussion	Textbook 1 Ch-1
2	M1	Cyber Kill Chain Methodology	PPT + case study	EC-Council
3		Lab Program 1: Introduction	Hands-on mapping exercise	Lab Manual
4		Lab Program 1: Cyber Kill Chain Simulation	Simulation & report	Lab Manual
5	M1	TTPs & Indicators of Compromise (IoCs)	Diagram-based teaching	Textbook 1
6	M1	Hacking Concepts & Hacker Classes	Interactive discussion	Textbook 2
7		Lab Program 2: Ethical Hacking Tools – Nmap	Tool-based practical	Nmap Guide
8		Lab Program 2: Ethical Hacking Tools – Metasploit	Hands-on demo	Metasploit Docs
9	M1	Phases of Hacking Cycle	Flowchart explanation	EC-Council

10	M1	Ethical Hacking Concepts & Scope	Lecture + Q&A	Textbook 2
11		Lab Program 3: Malware Analysis Basics	Sandbox analysis	VirusTotal
12		Lab Program 3: Malware Analysis (Contd.)	Practical + report	VirusTotal
13	M2	Password Security Fundamentals	Lecture	Textbook 2
14	M2	Password Complexity & Policies	Case discussion	Microsoft Docs
15		Lab Program 4: Password Cracking – John	Hands-on cracking	John Docs
16		Lab Program 4: Password Cracking – Hashcat	Practical session	Hashcat Wiki
17	M2	Password Attacks	PPT + attack flow	Textbook 1
18	M2	Countermeasures for Password Attacks	Discussion	OWASP
19		Lab Program 5: Social Engineering Simulation	Phishing simulation	OWASP
20		Lab Program 5: Awareness & Defense	Role-play + report	OWASP
21	M2	Social Engineering Techniques	Role play	Textbook 1
22	M2	Insider Threats & Identity Theft	Case study	CERT-IN
23	M2	Lab Program 6: Insider Threat Case Study	Scenario analysis	Course Notes
24	M2	Lab Program 6: Case Report & Presentation	Report submission	Course Notes

25	M3	Packet Sniffing Concepts	Lecture	Textbook 1
26	M3	Sniffing Countermeasures	Discussion	OWASP
27		Lab Program 7: Packet Sniffing	Wireshark capture	Wireshark Docs
28		Lab Program 7: Traffic Analysis	Analysis & reporting	Wireshark Docs
29	M3	DoS & DDoS Attacks	Lecture	CERT-IN
30	M3	Session Hijacking Concepts	Lecture	Textbook 2
31		Lab Program 8: DoS/DDoS Simulation	Attack + firewall rules	Lab manual
32		Lab Program 9: Session Hijacking Attack & Prevention	Session hijack demo	OWASP
33	M3	Web Server Architecture & Attacks	Lecture	Apache docs
34	M3	Web Application Vulnerabilities	Case studies	Web Hacker's Handbook
35		Lab Program 10: SQL Injection Testing	DVWA/BWAPP testing	OWASP
36		Lab Program 10: SQL Injection Mitigation	Secure coding	OWASP
37	M4	Wireless Security Fundamentals	Lecture	Textbook 1
38	M4	Wireless Encryption Algorithms	Discussion	IEEE
39		Lab Program 11: Wireless Attacks	Aircrack-ng demo	Tool Docs
40		Lab Program 11: Wireless Defense	WPA security config	Lab Manual

41	M4	Bluetooth Attacks	Lecture + demo	Mobile Security Refs
42	M4	Mobile Attack Vectors	Case study	OWASP Mobile
43		Lab Program 12: Mobile Security Tools	Tool demonstration	Tool docs
44		Lab Program 12: Mobile Attack Simulation	Practical	Tool docs
45	M4	IoT Security Overview	Lecture	Research Papers
46	M4	IoT Attacks & Tools	Case-based learning	Journals
47		Lab Program 13: IoT Vulnerability Testing	Hands-on	Lab Notes
48		Lab Program 13: IoT Security Analysis	Analysis & report	Lab Notes
49	M5	OT Security Concepts	LectureIndustry examples	NIST
50	M5	OT Attacks & Countermeasures	Lecture	Standards
51		Lab Program 14: OT Attack Simulation	Practical	Lab Manual
52		Lab Program 14: OT Defense	Hands-on	Lab Manual
53	M5	Cloud Computing Fundamentals	Lecture	AWS Docs
54	M5	Container Technology & Cloud Threats	Docker demo	Docker Docs
55		Lab Program 15: Cloud Security Configuration	AWS/Azure practice	Cloud Docs

56		Lab Program 15: Cloud Penetration Testing	Guided testing	CSA
57	M5	Penetration Testing Fundamentals	Lecture	Textbook 2
58	M5	Pen-Testing Methodology	Flow-based teaching	PTES
59		Lab: Hands-on Penetration Testing	Tool-based practice	Tool refs
60		Lab: Course Review & Viva	Evaluation & discussion	All References

Note: 1

2 credit courses	30 hours
3 credit courses	45 hours
4 credit courses	75 hours

Note: 2

- A course shall have a minimum of 3 modules and maximum 5 modules
- Each module will be covered in minimum 6 hours and maximum 12 hours

Assessment and Evaluation

Continuous Internal Evaluation (CIE)	
Components of CIE	Weightage of each component
CIE-1 - Quiz - Assignment	10 10
CIE-2 - Write-up (pen and paper)	25
CIE-3 - Tutorial and Lab report - Case-study Assignment and Presentation	10 15

Semester End Exams (SEE)	
Mode of Exam	Write-up
Weightage of exam	30%

Course Outcomes- Program Outcomes Matrix

Program Outcome Course Outcome		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO1	Understand the fundamental ethical hacking concepts and computer-based vulnerabilities to effectively identify and mitigate security risks.	3	2	-	-	1	2	-	2	-	3	1	-
CO2	Analyze various password cracking and social engineering techniques, along with their countermeasures, to enhance network and system security.	2	3	1	-	2	3	-	2	-	3	1	-
CO3	Analyze the various hacking options and security tools available in mobile and wireless	2	3	2	-	3	2	-	2	-	3	1	-

	networks to enhance network security.												
CO4	Analyze the different attacks and tools available in network-level, IoT, OT, web application and cloud computing.	2	3	2	2	3	2	-	3	-	3	1	-

Rubrics for evaluation of CIE (separate for each CIE component)

CIE Components		Marks	Rubrics
CIE-1	MCQ/MSQ	10	Multiple selection questions of 10 marks
	Assignment	10	Assignment will be evaluated for 10 marks based on assessment criteria prepared by the course faculty.
CIE-2	Pen and Paper	25	Test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.
CIE-3	Lab report	10	Tutorial and Lab execution reports need to be submitted in soft binding.
	Case-study Assignment & presentation	15	Prepare the research article related to the course and submit the draft copy and presentation.

CIE 1: Assignment (10 Marks)					
Sl. No.	Criteria	Measuring Methods	Excellent (8–10)	Good (5–7)	Poor (0–4)
1	Case Understanding & Scenario Analysis	Accuracy in identifying the incident, stakeholders, and forensic challenges	Demonstrates precise understanding of the case, identifies all key issues logically	Shows reasonable understanding with minor gaps	Limited understanding: key elements misunderstood or missing
2	Forensic Methodology & Process Application	Description of acquisition, preservation, analysis, documentation steps	Applies correct forensic procedures with strong justification and sequence	Applies appropriate methods but with limited justification	Incorrect, incomplete, or missing forensic steps

3	Quality of Evidence Interpretation & Findings	Accuracy of findings, linkage to evidence, clarity in reasoning	Interprets evidence accurately with clear, logical conclusions	Provides partially correct interpretation with some gaps	Incorrect, unclear, or unsupported interpretation of evidence
4	Presentation, Structure & Clarity	Report structure, organization, clarity, formatting	Well-structured, clear, well presented	Adequately structured; minor issues in clarity	Poorly structured, unclear, lacks professional format

CIE 3- Tutorial and Lab report					
Sl. No.	Criteria	Measuring Methods	Excellent (8–10)	Good (5–7)	Poor (0–4)
1	Execution of Experiment	Accuracy and completion of lab activity	Fully executed with correct results	Completed with minor errors	Incomplete or wrong results
2	Report Submission	Quality of report (Aim, procedure, results)	Clear, complete, well-structured	Adequate but missing minor details	Poorly documented or incomplete
3	Analysis & Interpretation	Quality of analysis and understanding of results	Insightful interpretation	Limited analysis	No analysis or wrong inference
4	Viva-voce	Q&A	Answer all queries with minor mistakes	Satisfactory	Poor

CIE 3: Case-study (15 Marks)					
Sl. No.	Criteria	Measuring Methods	Excellent (8–10)	Good (5–7)	Poor (0–4)
1	Problem Identification & Relevance	Clarity and originality of research problem	Innovative and well-defined topic	Relevant but limited scope	Unclear or repetitive topic
2	Content & Research Depth	Quality of literature, methodology, and findings	Comprehensive and well-supported	Adequate but lacks rigor	Poorly supported or incomplete

3	Execution	Accuracy and completion of execution	Fully executed with correct results	Completed with minor errors	Incomplete or wrong results
4	Presentation & Communication	Delivery, visuals, and ability to answer	Clear, confident, and engaging	Clear but lacks depth	Unclear, unprepared